Phase Diagrams: Reading the Map of Matter

Chemistry

Name: _____

Date: _____ **Period:** _____

1 OPENING QUESTIONS

Think about it — circle your gut instinct (no calculation needed):

1. Can water boil at only 70 °C? **Yes / No**
   Hint: think about what happens to a pot of water on a mountain where the atmosphere is thin.

2. Can water stay liquid at 200 °C? **Yes / No**
   Hint: a pressure cooker is a sealed pot that traps steam, raising the internal pressure well above normal atmospheric pressure. What might that do to the boiling point?

2 THE PHASE DIAGRAM

*Annotate **Diagram 1** below as your teacher builds it. Fill in the blanks on the diagram and in the definitions table.*

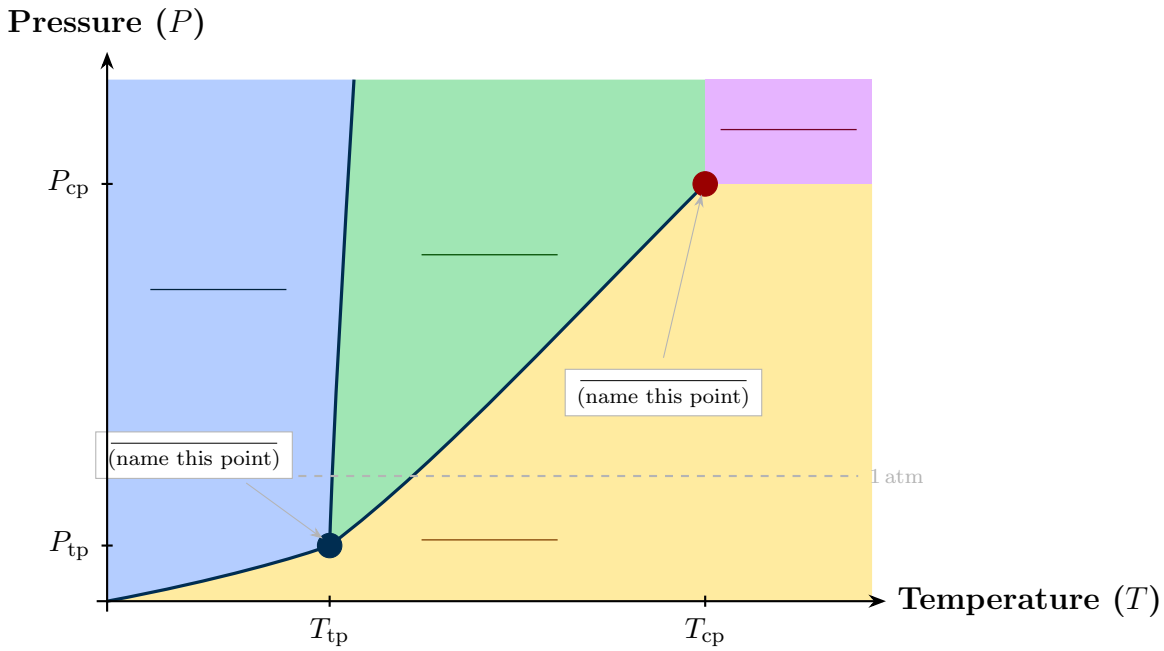


Diagram 1. *Generic phase diagram.*

Fill in the four blanked region labels and the two special points, and label the solid lines (ie phase boundaries).

New terms used on this diagram:

- **Vapor pressure:** the pressure exerted by a substance's gas phase when it is in equilibrium with its liquid (or solid) phase at a given temperature. The *Vapor Pressure Curve* shows how this pressure changes with temperature.
- **Sublimation:** a phase transition in which a solid converts *directly* to gas, skipping the liquid phase. Dry ice (solid CO_2) sublimes at normal atmospheric pressure.
- **Supercritical fluid:** a state of matter that exists when *both* temperature and pressure exceed the critical point values ($T > T_{cp}$ *and* $P > P_{cp}$). It has properties of both a liquid and a gas.

Key Features of Phase Diagram 1 — fill in as we go

Feature	What it means / represents
Each *region*	A range of (T, P) where the substance exists as one phase:
Each *boundary line*	A set of (T, P) values where:
Triple Point	The unique (T, P) where:
Critical Point	Above this (T, P),
Supercritical Fluid	

Connection to Heating Curves: On a heating curve, the flat plateaus represent phase changes where temperature stays constant as heat is added. Where do these flat regions appear on Diagram 1?

3 GUIDED PRACTICE

Work these problems with your teacher. Show all work.

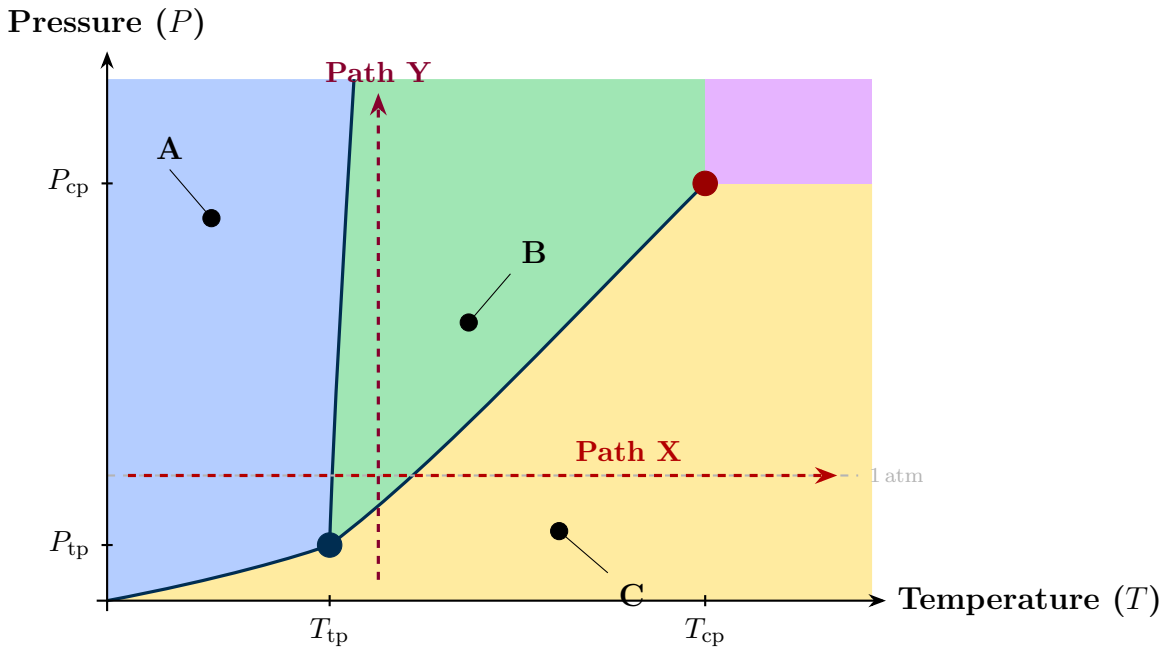


Diagram 2. *Generic phase diagram for practice problems.*

1. Identify the phase at each labeled point on Diagram 2.

   Point A: _____ Point B: _____ Point C: _____

   What is happening to the substance when it is exactly *on* the solid–liquid boundary?

   _____

2. Describe what happens as the substance travels along each path in Diagram 2. Name the phase transition(s) involved.

   Path X (horizontal: increasing temperature at constant pressure = 1 atm):

Path Y (vertical: increasing pressure at constant temperature):

3. The phase diagram of CO_2 differs from the generic one above: its triple point is at **5.1 atm** and **−56.6 °C**, meaning the entire 1 atm horizontal line sits *below* the triple point.

   (a) Does liquid CO_2 exist at atmospheric pressure (1 atm)? Explain using the concept of the triple point.

   (b) What phase transition does dry ice (solid CO_2) undergo when left out at room temperature and 1 atm? Name the transition.

   (c) What condition would you need to create in order to produce *liquid* CO_2?

Problem 4 — The Clausius–Clapeyron Equation

Recall that the **vapor pressure** of a liquid is the pressure exerted by its gas phase when the two are in equilibrium at a given temperature. As temperature rises, more molecules escape the liquid and the vapor pressure increases — this is exactly what the vapor pressure curve on the phase diagram shows. A liquid **boils** when its vapor pressure equals the surrounding atmospheric pressure.

The shape of the vapor pressure curve is not arbitrary — it obeys the **Clausius–Clapeyron equation**:

$$\ln\left(\frac{P_2}{P_1}\right) = -\frac{\Delta H_{\text{vap}}}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

where:

- P_1, P_2 = vapor pressures at temperatures T_1, T_2 (use any consistent pressure unit)
- T_1, T_2 = temperatures in **Kelvin** (always convert: $T(\text{K}) = T(^\circ\text{C}) + 273$)
- ΔH_{vap} = **molar enthalpy of vaporization**: the energy (in J/mol) required to vaporize one mole of liquid at constant pressure
- $R = 8.314\,\text{J/(mol·K)}$ = the ideal gas constant

For **water**: $\Delta H_{\text{vap}} = 40{,}700\,\text{J/mol}$.

At 100 °C (373 K), water's vapor pressure equals exactly 1.000 atm — this is the *definition* of the normal boiling point.

(a) At the summit of Mt. Everest, atmospheric pressure is approximately **0.330 atm**. Use the Clausius–Clapeyron equation to find the boiling point of water at this pressure.

Hint: set $P_1 = 1.000$ atm and $T_1 = 373$ K; $P_2 = 0.330$ atm; solve for T_2.

$T_2 =$ _____ K $=$ _____ °C

(b) Denver, CO (the "Mile High City") sits at ≈1609 m elevation, where atmospheric pressure is about **0.830 atm**. Without fully re-solving, predict qualitatively: will the boiling point of water be *higher* or *lower* than 100 °C? Then verify with a quick calculation.

$T_2 \approx$ _____ °C

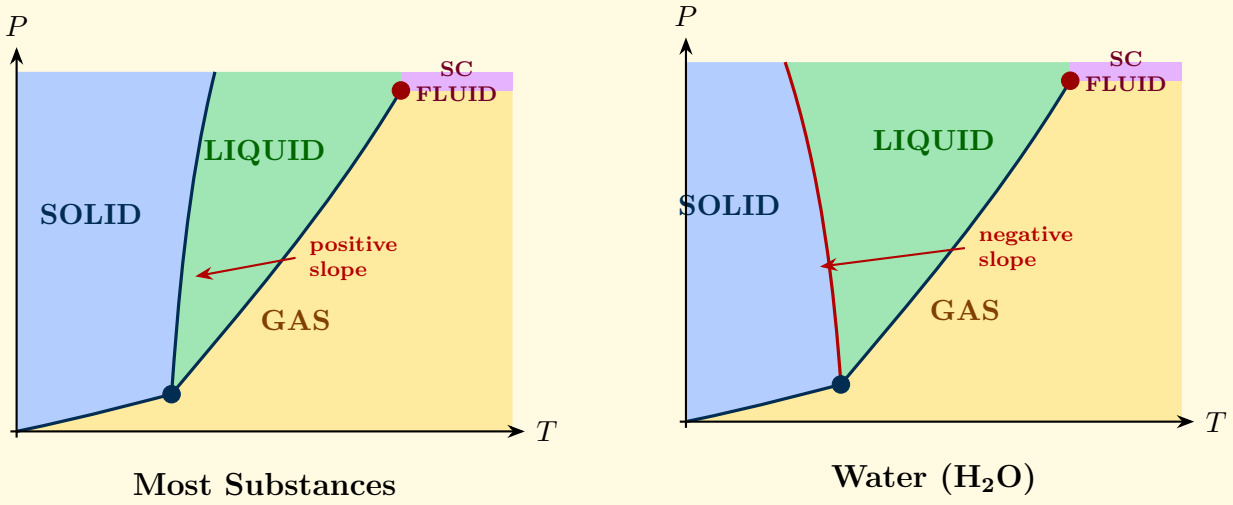
(c) Explain in one or two sentences why food takes *longer* to cook at high altitude, using your results from (a) and (b).

(d) **Challenge:** A pressure cooker raises internal pressure to approximately **2.00 atm**. Find the boiling point of water inside the pressure cooker. How does this relate to faster cooking?

$T_2 \approx$ _____ °C

4 WATER'S ANOMALY + IMF CONNECTION

Diagram 3 — Most Substances vs. Water



The red solid–liquid line on water's diagram slopes to the **left** as pressure increases. For most substances, it slopes to the right. This has significant physical consequences.

1. What does the **negative slope** of water's solid–liquid line mean physically? (Hint: starting in the solid region near 0 °C, which way do you move on water's diagram when you increase pressure? What phase do you end up in?)

2. You studied IMFs earlier in the year. Ice has a hydrogen-bonded crystal lattice that is *less dense* (more open) than liquid water. How does this explain the negative slope of water's solid–liquid line?

3. Ice skating is sometimes said to work because pressure from the blade melts a thin layer of ice. Does water's phase diagram *qualitatively* support this idea? What limitations does this explanation have?

Exit Reflection *Complete in 90 seconds.*

"A phase diagram is useful because . . ."